

What Is Cloud Computing Security?

Cloud computing security is a combination of controls, policies, and technologies used to protect environments, data, and applications deployed and maintained on the cloud. Cloud computing is the on-demand delivery of data storage and computing power.

It allows organizations to focus their resources on development and innovation while CSPs take care of setting up and scaling servers. It should be noted that cloud computing security is different from cloud-based security. Cloud-based security solutions generally come under security-as-a-service (SECaaS), offering centralized security services hosted on the cloud. They do not need dedicated resources in the company's existing infrastructure.

The cloud essentially functions on the internet, and cloud security refers to measures implemented to particularly address issues arising from cloud-based operations. When we access and store things on the cloud, we use remote servers with an internet connection instead of local hardware. It equates to renting physical space in a storage unit. [Cloud service providers \(CSPs\)](#) take care of allotting and maintaining this physical space for your assets.

As per the 2020 Cloud Computing Study by IDG, 81% of organizations host at least one application or module of their infrastructure on the cloud. This is up by 8% in just two years. The reasons why organizations are increasingly moving to the cloud are quick deployment, low set-up costs, geographical flexibility, and, most importantly, scalability.

Without cloud computing in the picture, most enterprises rely on firewalls and endpoint device management for security. Having in-house infrastructure reduces security concerns for certain aspects of the system, such as data storage. This is reflected in the fact that most security attacks with on-premise architecture are [malware-related](#), such as denial of service and sniffing attacks.

Types of cloud deployment

1. **Private cloud deployment:** A [private cloud](#) or internal cloud provides on-demand computing services for an internal network that can only be accessed by a single enterprise. The cloud is usually maintained by in-house IT staff. HPE, VMWare, and CISCO are some of the most popular private cloud providers.

2. **Public cloud deployment:** A [public cloud](#) refers to the computing services offered over the general internet. Third-party providers such as AWS and Microsoft Azure offer data centers and servers to customers by treating them as tenants. In this case, most of the control is with the provider. With public clouds, three types of services can be consumed.

- **Platform-as-a-service (PaaS):** These are services that provide the necessary infrastructure, plus a self-service model of managing and configuring all that it offers. It comes with predefined environments for development, testing, and active deployments. All that the application developers have to do is write and run the code. In this case, both the cloud provider and the organization take up responsibility regarding security. E.g., Heroku.

- **Infrastructure-as-a-service (IaaS):** These services provide basic infrastructure needs such as network, storage, and operating systems, on-demand. It allows for organizations to buy, build, and configure on their own. In this case, the organization takes up most of the security responsibilities. E.g., AWS.

- **Software-as-a-service (SaaS):** These are [software solutions](#) hosted on a cloud and delivered to the organization over a web browser or by using web interfaces such as APIs. This includes everyday applications such as GSuite and Salesforce. In this case, the organization availing the service is responsible for data storage and

user access. The provider takes care of everything else, such as encryption.

3. **Hybrid cloud deployment:** While it makes sense for highly sensitive assets to be deployed on a private cloud, it might prove too expensive to completely rely on private cloud-based infrastructure. To circumvent this issue, many organizations go in for [hybrid deployment](#). Sensitive and compliance-regulated workloads are run on private clouds, while less crucial operations are run on a public cloud. However, this system encounters challenges in the form of data migration.

As seen above, each type of deployment comes with a unique set of [security challenges](#). The inherent nature of a [cloud-based architecture](#) implies a shared security responsibility between the cloud service provider and the consumer.

This makes it more complicated than traditional on-premise setups.

With cloud computing, the perimeter of an attack is not as defined as traditional setups since it is similar to a moving fence. There is a constant movement of data between the enterprise system and both managed and unmanaged devices.

This means that security measures cannot be focused on just malware-related attacks. It involves multiple layers of firewalls—one for external communication and another for internal management.

With CSPs providing dedicated security measures as part of their SLAs, data breaches become less likely. However, some vulnerabilities of a cloud-based infrastructure include side-channel attacks, port and [vulnerability scanning](#), account takeovers, and ransomware. In the coming section, we shall discuss the key risks associated with cloud computing, despite its varied advantages.

Need for Security: Key Risks Associated With Cloud Computing

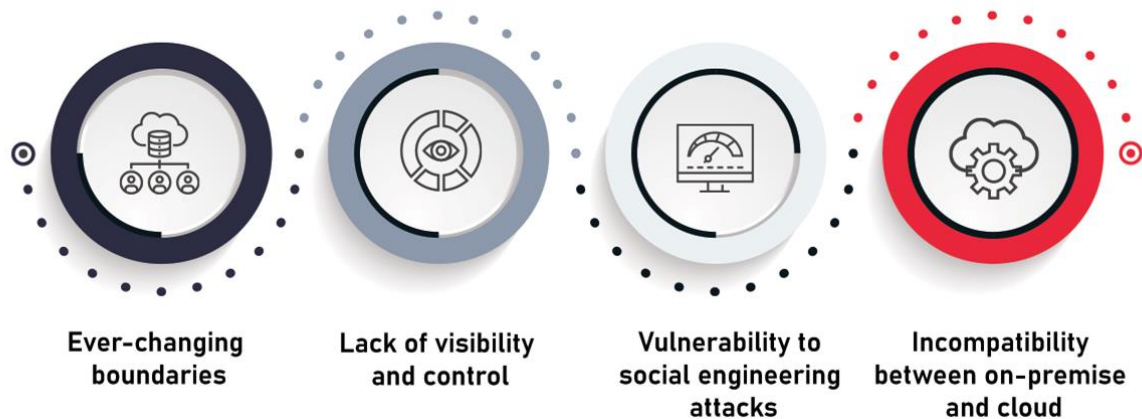
The future of enterprise applications is touted to be deeply entangled with cloud computing, and current statistics reflect that. McAfee's 2019 Cloud Adoption and Risk Report shows that an average enterprise uses around 1,935 cloud services. Cloud computing addresses our rapidly changing, geographically distributed technology scene. It also addresses the needs of smaller firms and startups that do not have the capital to invest in their own infrastructure.

We touched upon 'shared responsibility' in the previous section, and that plays a vital role in how secure an organization's cloud can be. Cloud service providers are bound by SLAs to provide safe housing for their custom applications with failover mechanisms in place. This means that the onus of complete security falls on the organization itself.

The [cloud security alliance \(CSA\)](#) annually examines the inherent risks of cloud computation and releases a report. The last report focused on configuration and authentication issues instead of its traditional malware and vulnerability focus. With increasing amounts of sensitive data being housed on the cloud, these risks become crucial to business continuity.

Listed here are the risks that come with moving infrastructure fully or even partially to a cloud setup.

RISKS ASSOCIATED WITH CLOUD COMPUTING



1. Ever-changing boundaries

Before cloud computing officially arrived, network perimeters were well defined. Any changes to these boundaries were slow and within the company's control. Cloud environments, however, do not have a well-defined boundary. The tested cyber-security tactics of perimeter defense are therefore not sufficient. With cloud computing, the question of security is no longer constricted to blocking hackers at the fence. Security must equally focus on keeping data safe.

2. Lack of visibility and control

CSPs do not expose all details and full control of the [infrastructure](#) to their customers. This means that the admin team of the client company does not have all the information it possessed with an on-premise-only setup. In such a case, it becomes difficult to identify and visualize all cloud assets.

3. Workload distribution

One of the best things about using a CSP is the on-demand provisioning and decommissioning of resources. This means that the work environment is

constantly shifting. It also means that the infrastructure is scaled up at a very rapid pace. A cloud security strategy is crucial to address this happy problem to prevent security holes. All security policies must apply uniformly across different clouds and on-premise segments.

4. Compliance challenges

[Cloud compliance](#) is important since laws such as GDPR that govern how and where data is stored are coming into effect everywhere. Any misstep in this direction can lead to legal and financial ramifications. Organizations must address important questions in the cloud service provider's SLAs. These questions can vary from 'How is the data segregated from other organizations?' to 'Do you have an [incident response plan](#) in place?'.

5. DevOps, CI/CD, and shadow IT

Most organizations use continuous integration/continuous delivery (CI/CD) for constant application updates with minimal downtime. CI/CD with the cloud requires extra measures such as instant and automatic scaling, deployment across varied clouds and virtual machines (VMs), and putting in checkpoints and responsible gatekeepers for every module of the application.

Shadow IT is the use of applications or devices without official approval from the IT admin. Cloud computing enables easy procurement of services by anyone in the organization, circumventing the procurement policies laid down by the IT department, and this affects the application stack. The varied deployment and maintenance requirements that come with using different cloud services also mean that DevOps members resort to using multiple, non-sanctioned local IT tools to ease operation.

6. Misconfiguration or no configuration

According to the 2019 McAfee Cloud Adoption and Risk Report, enterprises experience an average of 2,269 individual misconfiguration-related incidents per month. They also found that 1 in 20 AWS S3 buckets (Amazon's cloud storage buckets) have read permissions enabled for everyone to consume. These issues occur due to improper configuration from the organization's end or just sticking to the default setting provided by the CSP.

7. Complexities in data transfer

Using cloud-based services involves constant data transfer across networks. This is usually done using web services. For the secure movement of data through these services, organizations need to look into encryption, access controls, authorization, and authentication.

8. Access control and privileges

By default, cloud service providers usually give very broad access permissions, especially with SaaS applications. If user roles and privileges are not properly defined, it leaves the organization exposed to cyberattacks. For example, not every employee requires to have the right to delete assets on the cloud. Any consideration of a cloud-based architecture must be accompanied by implementing a strong [identity and access management \(IAM\)](#) plan.

9. Increased vulnerability to social engineering attacks

Having assets on a public cloud opens up the organization's system to hacking and [social engineering attacks](#). If employees do not go through security

awareness training, they might fall prey to phishing attempts. One example of a major incident triggered by a [phishing](#) attempt is the 2020 Twitter bitcoin scam.

10. Incompatibility between on-premise and cloud

We've already highlighted the differences in security strategies between being completely on-premise and moving certain components to the cloud.

Unfortunately, this means that traditional on-premise tools may not address all security risks. As a result, organizations need to ensure they have the relevant expertise and tools to spot the visibility and control gaps during the transition.

Top 10 Best Practices for Cloud Computing Security

As with every other technology, there are risks in moving to cloud computing. However, this does not mean that one can just throw away the notion of the cloud, along with its varied benefits. To counter the risks mentioned above, organizations must ensure that they follow the best practices prescribed by industry experts and organizations such as CSA.

TOP 5 BEST PRACTICES FOR CLOUD COMPUTING SECURITY



1. Segment and isolate the system

The multi-tenant setup of the public cloud means extra attention must be given to segmenting the system. The best way of doing this is by dividing the system into zones. Each zone is used to isolate instances, containers, applications, and corresponding data stores. An intelligent segmentation strategy can be key to ensuring that even if one component goes down, the entire system does not crash.

2. Ensure identity access management (IAM) hygiene

Identity and access management (IAM) is crucial to block security problems created by malicious intent (hackers) or even plain negligence (insider threats). Minimal access must be granted to critical assets and web services used to transfer data. The more privileges granted to a user, the higher the level of required authentication. This is where multi-factor authentication (MFA) comes into place. Privileges must be role-based, and all access privileges must be constantly audited and revised. In addition to access policies, good IAM

hygiene, such as strong password policies and permission timeouts, must also be implemented.

3. Maintain proper lifecycles

With such ease of creation in the cloud, it is easy to spin new instances and abandon old ones. Unfortunately, these abandoned instances are often neglected, with zero monitoring even though they are active. This means that the typical maintenance routines that run on active servers, such as applying security patches, may not happen on these orphan resources.

That is why it is important to come up with a lifecycle management strategy. [Documented policies](#) on configuration and access while commissioning and decommissioning instances can safeguard organizations from numerous vulnerabilities. [Governance and compliance policies](#) may also specify auditing configuration differences and remediating them automatically when incidents occur. This kind of cloud security automation is called [cloud security posture management \(CSPM\)](#). These kinds of policies and CSPM leave little room for shadow IT practices, which is a big plus.

4. Configure beyond the default

As mentioned above, CSPs are very generous with their default configuration. Retaining these opens up a lot of vulnerabilities. At least in the initial days of cloud adoption, it is best for organizations to rope in experts who can closely collaborate with the cloud providers to bring in optimal configuration policies. As with everything else, this must be audited and revisited regularly.

5. Perform vulnerability scans regularly

[Vulnerability management](#) is a big part of cloud computing security. Security audits must be thorough and regular. Every instance on the cloud must follow

the prescribed lifecycle, with security patches applied as per schedule. Most importantly, vulnerability scans must be performed.

Vulnerability scanners are tools that comb through the infrastructure to proactively spot potential threats and security holes. Vulnerability scans can only identify problems. The security team must quickly carry out remedial actions to prevent hackers from cashing in. It should be noted that these scans aren't just for maintaining good security hygiene. Compliance regulations that cover data privacy, such as NIST and HIPAA, emphasize vulnerability scanning to protect sensitive customer data.

6. Implement backup and recovery policies

[Disaster recovery plans \(DRP\)](#) are crucial for business continuity. While signing on with a cloud service provider, all questions pertaining to data backup, retention, and recovery policies must be asked and answered. It is important that these align with the internal standards set by the DRP team. This information must be used to come up with break-glass strategies.

It is also important that internally, the organization has backup plans for essential assets and data. This can be on another cloud or on-premise. An outage of the primary cloud service should not result in the entire system going down. That is why a failover plan is necessary.

7. Establish monitoring and alerting policies

All user activity must be constantly monitored across all environments and instances. [Access privileges](#) must be recorded using session monitoring. Every log from every instance must be centralized with appropriate reports generated to make things easier for the security team. A unified cloud management platform usually does the trick.

8. Carry out penetration testing

Cloud [penetration testing](#) is defined as the testing of a cloud-hosted system by simulating cyber attacks. It is used to assess the security posture of the system by identifying various strengths and weaknesses. In addition, penetration testing gives a good idea of the attack surface that the security team needs to work on. Besides the obvious advantages of identifying threats, penetration testing also allows the organization to identify different components of its infrastructure and how each is positioned. Penetration testing answers questions such as:

- Which systems are exposed to the public?
- Who has access to different services?

9. Implement consistent security policies

While it is tempting to implement different types of security measures for new cloud-based additions to the system, one of the best security decisions is ensuring consistent security across all clouds and data centers. This covers all infrastructure, including on-premise components.

Some basic security policies include:

- Creating user groups with clearly prescribed roles and privileges. This should directly map the different services the organization consumes and deploys. Multi-factor authentication (MFA) must be used where necessary. Most CSPs provide this feature, but it is always safe to confirm the same before subscribing to their services.
- Using [firewalls](#) to inspect and control traffic at each endpoint with automated firewall policy changes based on fluctuations in web traffic patterns is a huge security advantage. Firewalls can also be applied to microservices that run critical workloads. Firewall policies must factor in the ports that need to be open and the endpoints that need strict restrictions.

- Encryption at all transport layers helps organizations maintain good data storage hygiene as well as lowers compliance risks. Most CSPs provide encrypted services that range from encrypted connections to sensitive data encryption. It is up to the organization to figure out what sort of data protection is in place by default and what it needs to put in. For example, data might need to be encrypted with third-party software before upload if the cloud does not automatically do it.

[Cloud access security brokers \(CASBs\)](#) are the go-to for encryption-related policy execution. Encryption policies must cover both data at rest and data in transit.

10. Understand and adhere to compliance regulations

Cloud compliance is complex. It may be the same set of regulations as for basic on-premise solutions but needs to take into account the following:

- Hybrid networks: These combine public, private, and on-premise components.
- Multi-cloud components: When the architecture includes IaaS from multiple vendors.

Big cloud players such as AWS and Azure come with compliance-based certifications, such as for PCI DSS and HIPAA. However, this does not mean that they are solely responsible for maintaining compliance. CSPs provide control to the organizations they cater to, keeping in mind the hybrid infrastructure they may be dealing with.

In conclusion

In order to maintain a good cloud security posture, end-to-end visibility of the (often hybrid) system is essential. Security policies must be maintained and deployed across all components—the more automation here, the better.

[Cloud security](#) policy management services are available, and the market for complete cloud security services is teeming with many intelligent offerings.

Organizations can opt for in-house teams or one of these services. That being said, regular employee training is crucial to make cloud integration work.

Remember, the keywords here are ‘shared responsibility’.

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